

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.6816163525995613, median 0.43036124485195276, std: 0.8809350630089853
Gyroscope error (imu0): mean 3.95676488901969, median 2.0513803595271547, std: 8.084245308594356
Accelerometer error (imu0): mean 9.642060739793411, median 3.3484398587274917, std: 27.034153636161093

Residuals

Reprojection error (cam0) [px]: mean 0.6816163525995613, median 0.43036124485195276, std: 0.8809350630089853
Gyroscope error (imu0) [rad/s]: mean 0.0022697684412894383, median 0.0011767589259743217, std: 0.004637466563659237
Accelerometer error (imu0) [m/s^2]: mean 0.06831534970386277, median 0.02372416499797144, std: 0.1915407618187094

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.02262516 -0.99741723 -0.06816863 0.03543435]
[0.03018008 0.06747359 -0.9972645 0.07566829]
[0.99928838 -0.0246206 0.02857553 -0.12164531]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.02262516 0.03018008 0.99928838 0.12007677]
[-0.99741723 0.06747359 -0.0246206 0.02724224]
[-0.06816863 -0.9972645 0.02857553 0.08135289]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.07788917746099655

Gravity vector in target coords: [m/s^2]

[0.0524885 1.62565749 -9.67072415]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [158.84236747881792, 159.1388840108544]

Principal point: [159.39007244740083, 159.66667038693473]

Distortion model: radtan

Distortion coefficients: [-0.0029665211737014835, -0.0005195926033502311, 0.0005793199746489879, -0.002086187730906022]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

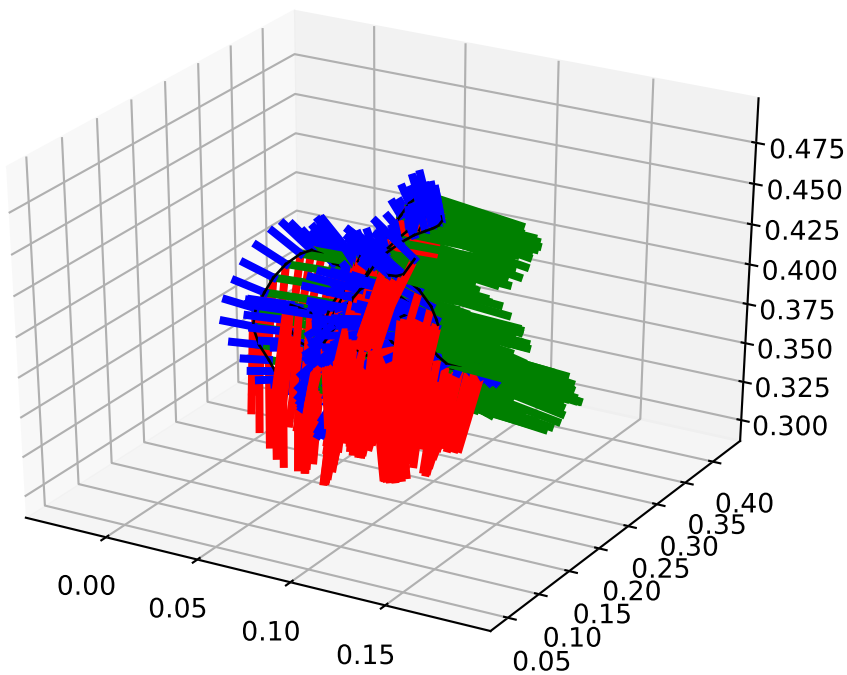
[0. 1. 0. 0.]

[0. 0. 1. 0.]

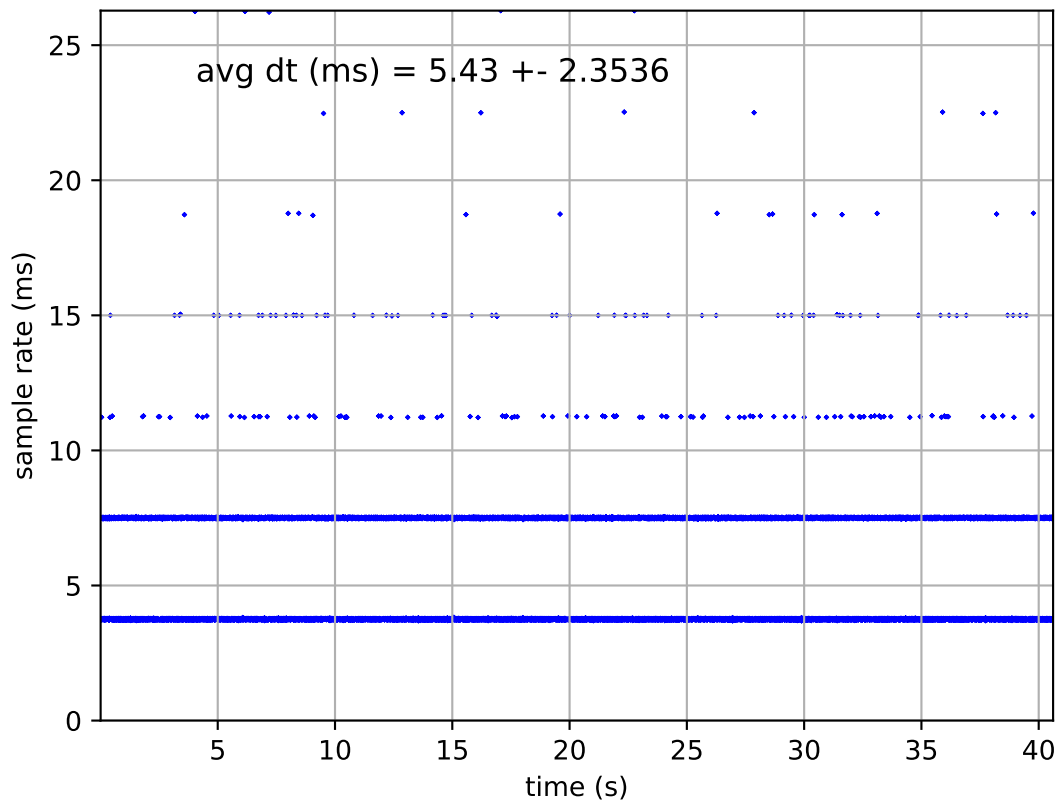
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

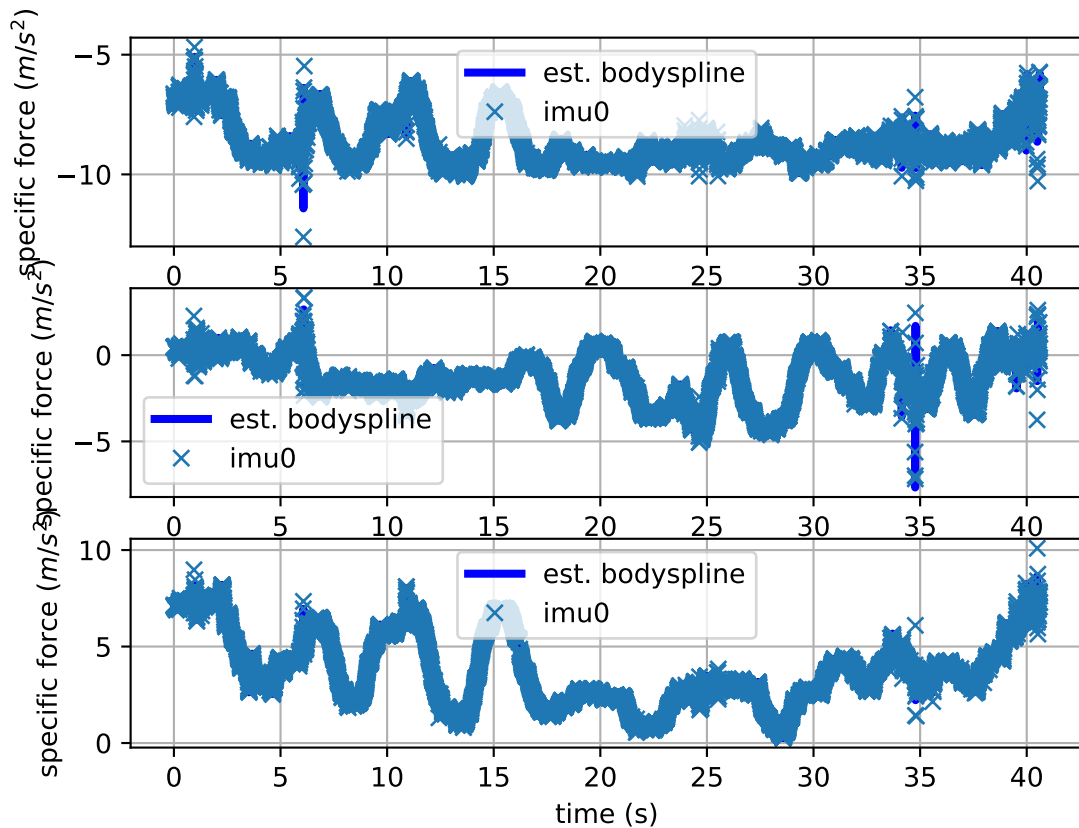
imu0: estimated poses



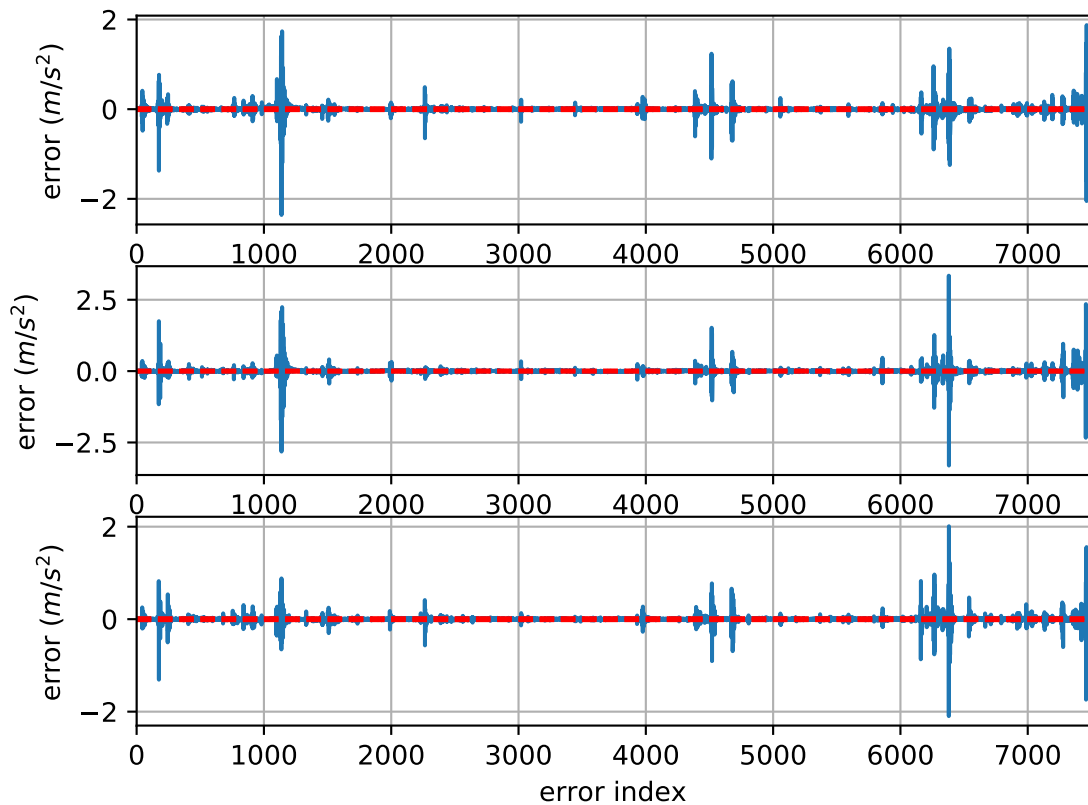
imu0: sample inertial rate



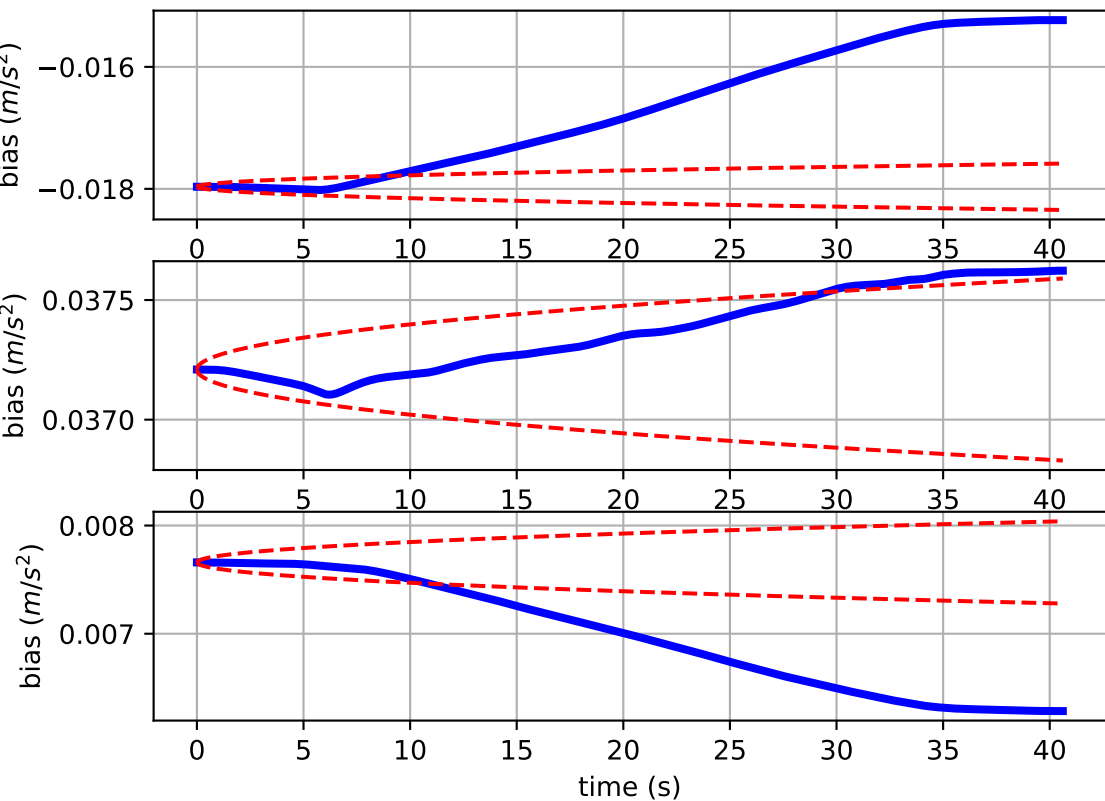
Comparison of predicted and measured specific force (imu0 frame)



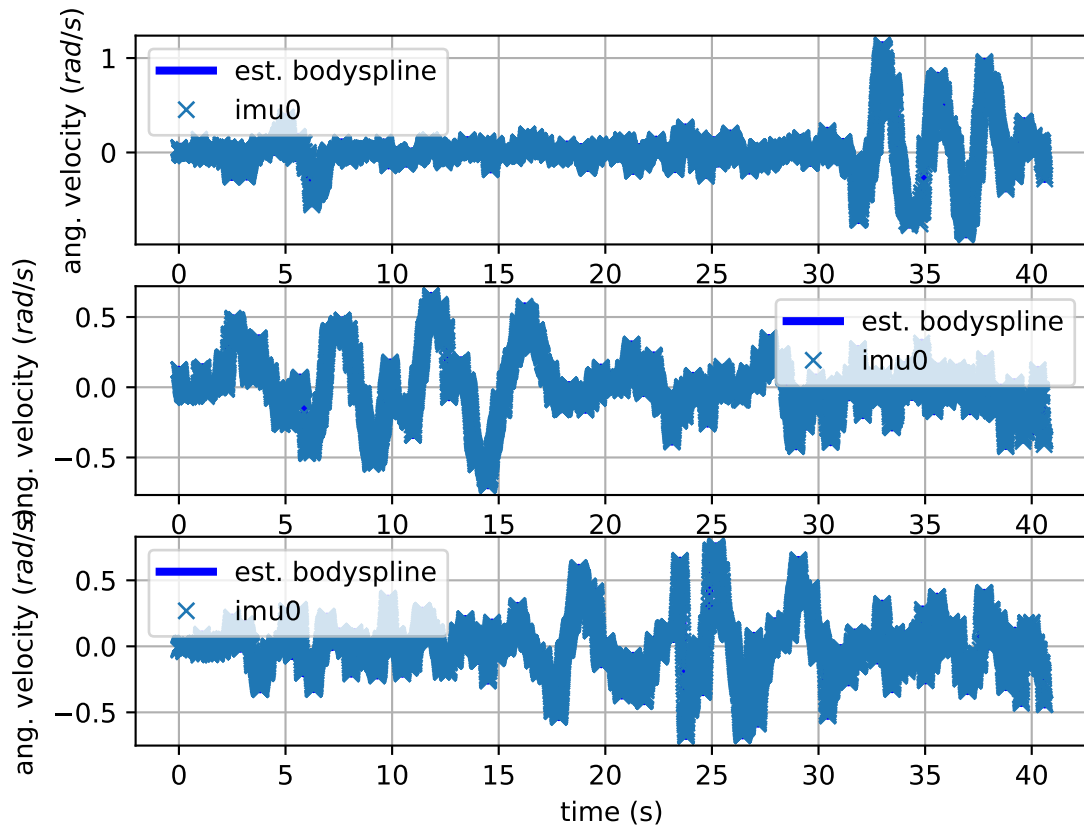
imu0: acceleration error



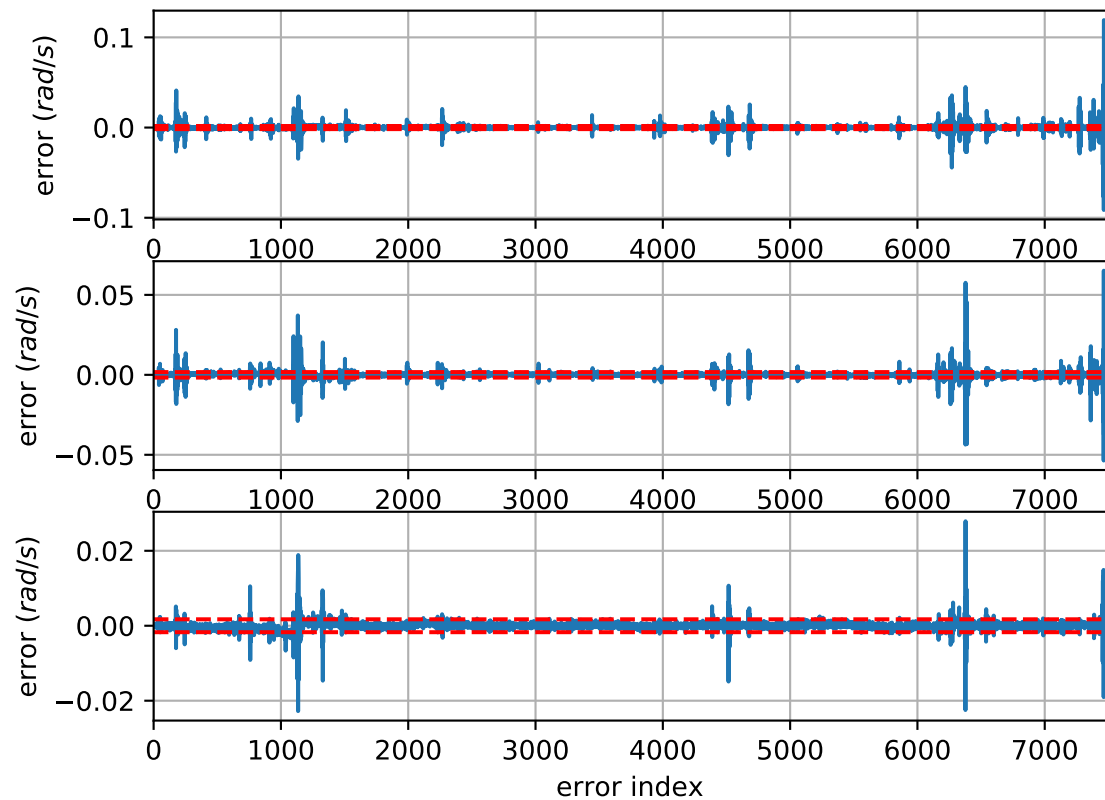
imu0: estimated accelerometer bias (imu frame)



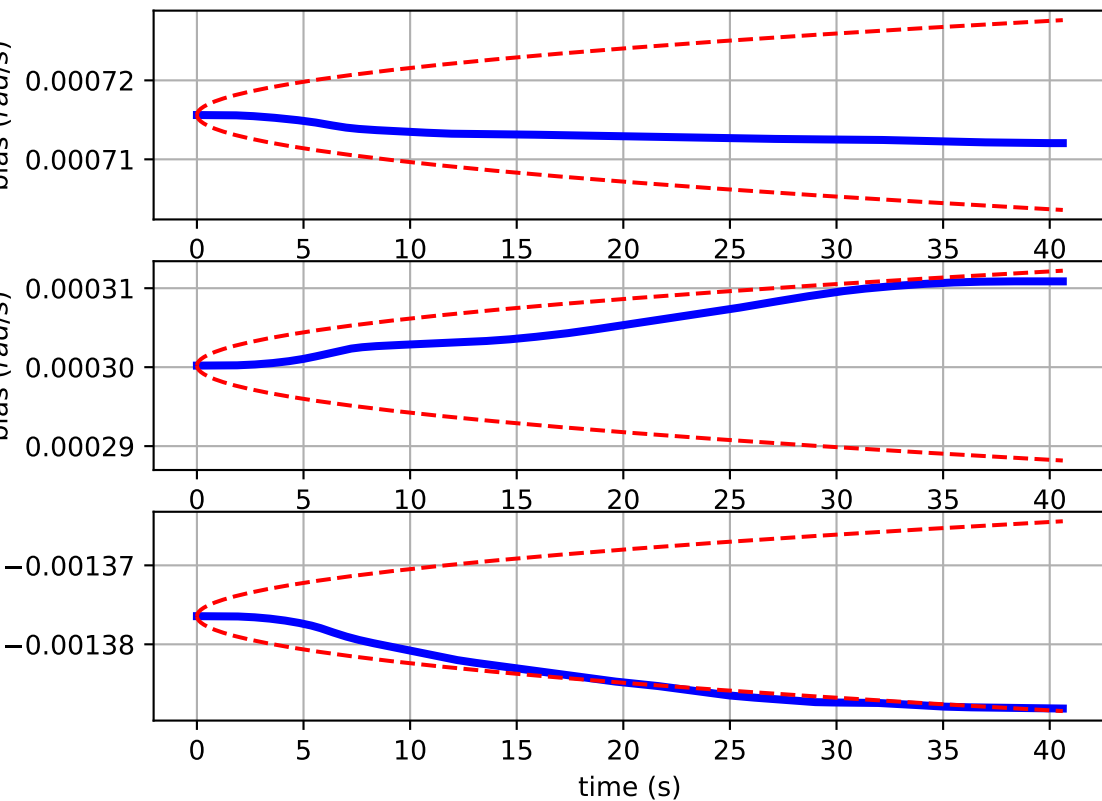
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

